

IT_DatabaseAdminUtility Data Migration Process

Introduction

This outline will detail and list the steps, process and methodology used when executing a migration using the Data Migration Utility on the Utility Server currently DEV-NY (hdw-app013583).

Utility Server System Directories

The solution will use a combination of FileMaker System scripts in tandem with OS level system scripts to perform the operations required to create/execute jobs that will copy, unzip, create directories, and execute the FMDM command line tool. The process will reference and rely on the ".bat" scripts currently located here:

F:\FileMakerRelease\scripts

There is no need to modify these files. The path to this directory can be updated in the Admin section of the solution.

The OS level system scripts are:

FMDM.bat – FileMaker Data Migration command line tool. Expects 6 parameters

Mkdir.bat – System script to create folders using mkdir command. Expects 3 parameters.

UnZipFile.bat – System script to unzip a file using TAR command. Expects 4 parameters.

Xcopy.bat – System script to copy a file using xcopy command. Expects 3 parameters

The FileMaker System scripts will be created dynamically, executed, then deleted upon completion. These FM system scripts will contain specific parameters to the process and will only be used for the migration process.

Utility Server Folder Migration Directories

Upon execution the Utility server will create a new host folder in the following location:

F:\FileMakerRelease\Databases\[DatabaseFolderName]\[YYYY-MM-DD]

Example: F:\FileMakerRelease\Databases\FIN_ProductionFinance\2024-07-30_0324AM

Additionally, within the host folder there will be three subfolders created:

- 1) "clone" – Directory will ultimately contain a copy of the file without data accompanied with a text log file. The FM file is the schema of the file that contains the source code from the environment selected. This file will be created using a FM Schedule to create the CLONE file and a system script job to copy the file from either DEV or BETA to this directory. The file will be the database filename and will append a space and the word "Clone" to it's file name.
 - a. "[DatabaseFileName] Clone.fmp12" eg. "FIN_ProdCostReporting Clone.fmp12"
 - b. "XcopyJobResult.txt" – Contains a log of the xcopy system script result when it copied the file from DEV or BETA to this directory.
- 2) "source" – Directory will contain a copy of the file used for the data source of the migration and will also contain a text log file. This file will be obtained from either a backup copy from PROD or when migrating to PROD, the hosted file from PROD. When the migration is to PROD, this folder will also contain a ZIP file of the target file. This ZIP file is created when the database is downloaded from the Server. The file with the '.fmp12' extension is the extracted file from the ZIP file. This file can be seen as a Backup to the file being migrated/replaced.
 - a. [DatabaseFileName.fmp12] eg. FIN_ProdCostReporting.fmp12
 - b. "XcopyJobResult.txt" – Contains a log of the xcopy system script when it copied the file to this directory.

- 3) "migrated" – Directory will contain the migrated file created by the FMDM command line tool and two text files that will contain details of the system scripts process. The .fmp12 file is the file that will be 'uploaded' to the Target machine and represents the payload of the process combining the schema from the CLONE directory and merging it with the data from the file in the SOURCE file.
- a. [DatabaseFileName.fmp12] eg. FIN_ProdCostReporting.fmp12
 - b. "XcopyJobResult.txt" – Contains a log of the xcopy system script when it copied the file to this directory.
 - c. "[Databasefilename]-FMDM-Result.txt" example: FIN_ProdCostReporting-FMDM-Result.txt

Technical Steps Overview

1) Environment Test/Validate

- A. Test
 - a. Test server communication to the CLONE and Target server with ADMIN account provided.
- B. Validate
 - a. Validate the file selected exists on the CLONE server
 - b. Validate the file exists on the Target server when migrating to PROD
 - i. When migrating to PROD if the file does not exist the process will abort.
 - c. Validate there are 0 users connected to the file on the Target server (BETA or PROD).

NOTE: When migrating from DEV to BETA the file does not need to already exist on the BETA server to continue.

2) Create Migration Folder Structure

- a. Folder Creation – Using FM Server system scripts and OS level system scripts create the folder structured outline above.

3) CLONE File Source

- a. Create / execute FM System Script job to create a CLONE job on either the DEV or BETA server.
- b. Create / execute FM System script job to execute OS level system script to copy the file to the Utility server CLONE directory.

4) Data SOURCE file

1. When migrating from DEV to BETA. Data source will be from a backup copy from the PROD server the file is hosted on currently formatted as: "Daily at 1am_[YYYY-MM-DD]_0100" eg. \\hdw-app013583\Backup\Daily at 1am_2024-07-15_0100\APP_DATA\[FolderName]\[DatabaseFileName.fmp12]
 - a. Create / execute FM System Script job to execute a OS level xcopy folder job.

Example: F:\FileMakerRelease\Databases\FIN_ProductionFinance\2024-07-30_0324AM\clone

Example: F:\FileMakerRelease\Databases\FIN_ProductionFinance\2024-07-30_0324AM\source

Example: F:\FileMakerRelease\Databases\FIN_ProductionFinance\2024-07-30_0324AM\migrated
2. When migrating from BETA to PROD. Data source will be the file currently hosted on PROD.
 - a. Download the file from PROD and save it in the Source directory as a ZIP file.
 - b. UnZip the file and place content of the file at the root level of the source directory.

5) FM Data Migration Tool

Create / execute FM System Script job to run the FMDM migration tool, referencing the files in the clone and source directory and creating the new file that will be used to upload to the target server.

6) Upload Migrated file to Target Server

a.) Remove file from the Target Machine

b) Upload file created from the FMDM migration tool to the Target server. File will be automatically opened upon completion of the upload.